

Directed graphs

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Graphs & Social Networks
Lecture 4

Outline



- 1 Basic concepts
- 2 Tournaments
- 3 Partially ordered sets

Outline



1 Basic concepts

2 Tournaments

3 Partially ordered sets



Reminder: graph connectivity

Definition

Vertices $u, v \in V$ of undirected graph G are **connected** if there exists uv -path in G , that means a path $v_0, \dots, v_k$ such that $v_0 = u, v_k = v$. A graph with pairwise connected vertices is called **connected graph**.

Definition

Each maximal (thus induced) connected subgraph of graph G is called its **connected component**.

Two weeks ago: Eulerian tours in digraphs



Theorem (Euler, 1736)

Finite and connected multigraph has Eulerian tour if and only if each its vertex has even degree.

How does it work for a directed case?



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Theorem

*Finite and **connected** directed multigraph has Eulerian tour if and only if for each its vertex v there is $\deg^+(v) = \deg^-(v)$.*



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What does it mean *connected* for directed graphs?



Two weeks ago: Eulerian tours in digraphs

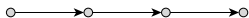
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What does it mean *connected* for directed graphs?



Is the graph on the left *connected*?

Question



How to define connectivity for directed graphs?

Directed & undirected

Definition

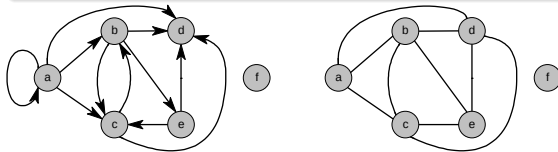
Let $G = (V, A, f)$ be a directed multigraph. The **undirected version** (or the **skeleton**) of graph G is undirected graph $G' = (V, E)$, such that

$$E = \left\{ \{u, v\} : u \neq v \wedge \exists a \in A f(a) \in \{(u, v), (v, u)\} \right\}$$

Definition

Let $G = (V, E)$ be a (simple) undirected graph. The **directed version** of graph G is the directed graph $G' = (V, A)$, such that

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And the third one :-)



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Lemma

Let $G = (V, E)$ be any undirected graph, and $G' = (V, A)$ — its directed version. Then the graph G is the skeleton of G' .



Connectivity — directed graphs

Definition

Directed graph G is **weakly connected** when its skeleton is a connected graph.

Definition

Let G be a directed graph and $u, v \in V(G)$. We say that vertex v is **reachable** from vertex u if there exists an uv -path in G , that means a directed path $v_0, \dots, v_k$ such that $v_0 = u, v_k = v$. A digraph in which the relation of being reachable is the total one (i.e equal $V(G) \times V(G)$) is called a **strongly connected** graph.

Strongly connected components. The structure of weakly connected digraph.



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Strongly connected components. The structure of weakly connected digraph.

Directed acyclic graphs



Definition

A digraph without (directed) cycles is called a **dag** (Directed Acyclic Graph).

Proposition

Digraph G is a dag, iff G has no loops and its strongly connected components are single-vertex.

A picture

Outline



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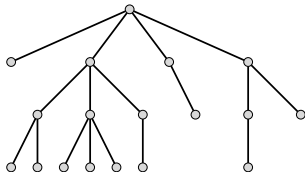
Digression — a different way of making arrows

Definition

Let $G = (V, E)$ be a simple directed graph. An **orientation** of G is a digraph $D = (V, E')$ obtained from G by choosing an orientation (x, y) or (y, x) for each edge $xy \in E$, $x, y \in V$.

A **tournament** is an orientation of a complete graph.

Rooted tree.





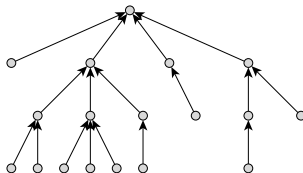
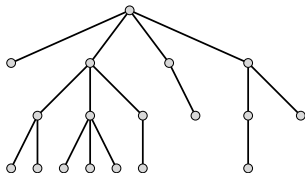
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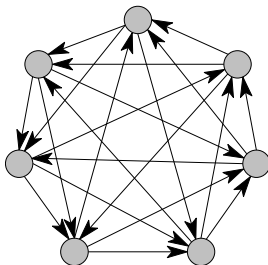
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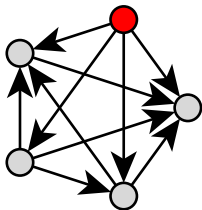


Tournament result

An outdegree $\longleftrightarrow$ a tournament score.

Extra exercise

Let $p_1, p_2, \dots, p_n$ be a finite sequence of natural numbers, $0 \leq p_1 \leq p_2 \leq \dots \leq p_n$. Let $p'_k = \sum_{i=1}^k p_i$. Prove that a tournament with outdegree $p_1, p_2, \dots, p_n$ exists if and only if $p'_k \geq \binom{k}{2}$ for $1 \leq k \leq n$ and $p'_n = \binom{n}{2}$.



It is not always possible to determine the winner of a tournament. Moreover, for any $k \in \mathbb{N}$ there exist k -paradoxical tournaments, i.e. for any $S \subseteq V(T)$, $|S| \leq k$ there exists a vertex $v_0 \in V(T) \setminus S$ such that $v_0 \rightarrow v$ for all $v \in S$.

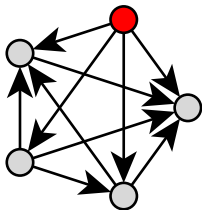


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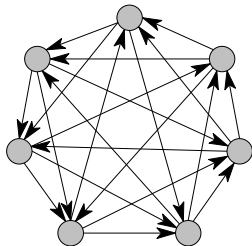
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The king of tournament (a moral winner)

Definition

In a digraph a **king** is a vertex from which every other vertex is reachable by a path of length at most 2.

Proposition (Landau [1953]))

Every tournament has a king.

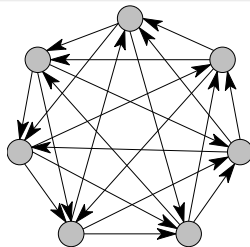
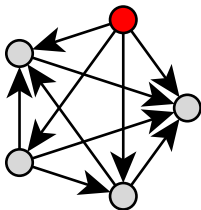
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Exercise

Let T be a tournament where everyone loses something ($\delta^-(T) > 0$).

- 1 Prove that given king x , there exists another king $y \in V(T)$ such that $y \rightarrow x$.
- 2 Prove (using (1)) that T has at least 3 kings.

There exists a n -vertex tournament with k kings for $n \geq k \geq 1$ with exceptions $k = 2$ and $n = k = 4$. (Maurer [1980]).

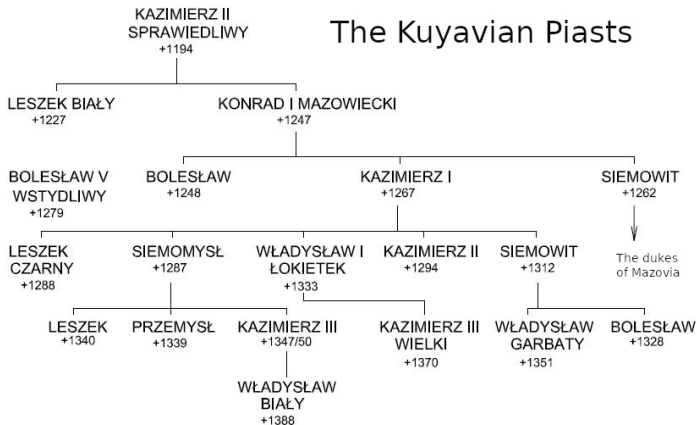
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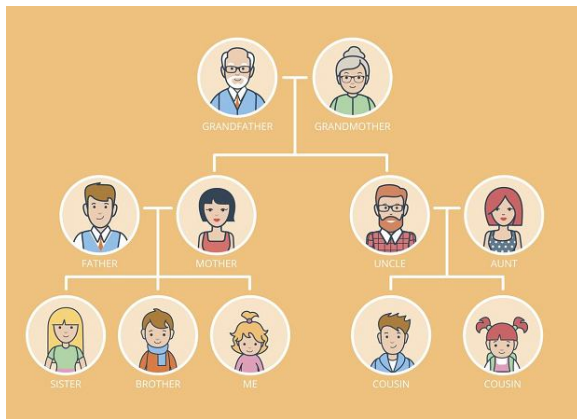
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Last lecture: family trees (genealogy)

The Kuyavian Piasts



Last lecture: family trees (?) (genealogy)





Types of homogeneous relation — once again

Definition

Let $R \subseteq X \times X$ be homogeneous relation. We say that R is:

- reflexive, when $\forall x \in X \quad xRx$;
- irreflexive, when $\forall x \in X \quad \neg(xRx)$;
- transitive, when $\forall x, y, z \in X \quad (xRy \wedge yRz) \Rightarrow xRz$;
- antisymmetric, when $\forall x, y \in X \quad (xRy \wedge yRx) \Rightarrow x = y$;
- connected, when $\forall x, y \in X \quad x = y \vee xRy \vee yRx$.

Relation $R \subseteq X \times X$ is a:

- **partial order**, when R is reflexive, transitive and antisymmetric;
- **strict order**, when R is irreflexive and transitive;
- **total order** (resp. strict), when it is furthermore connected
- relation of a tournament, when ... (?)



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- **total order** (resp. strict), (or **linear order**) when it is furthermore connected



Partially ordered sets

Definition

Let V be a set and $\preceq \subseteq V \times V$ — a partial order on V . A pair (digraph) $(V, \preceq)$ is called **partially ordered set** or **poset**.

Observation (Trivial fact)

For any partial order $\preceq \subseteq V \times V$ there exists a strict order $\prec \subseteq V \times V$ such that

$$\preceq = \prec \cup \{(x, x) : x \in V\} \text{ and } \prec = \preceq \setminus \{(x, x) : x \in V\}.$$

Maximal, minimal elements, a maximum, a minimum.

Question



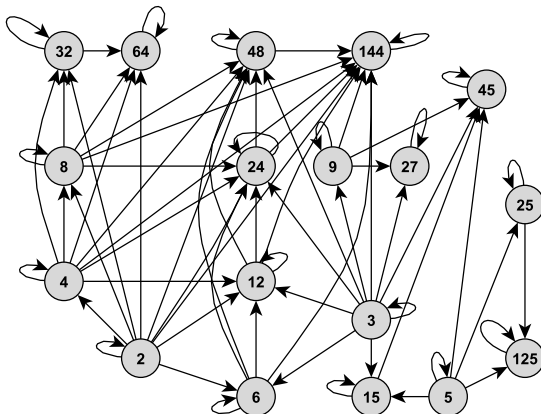
How to draw a poset legibly?

Question

How to draw a poset legibly?

$X = \{2, 3, 4, 5, 6, 8, 9, 12, 15, 24, 25, 27, 32, 45, 48, 64, 125, 144\}$.

$\preceq$ means
divisibility
on X .



Transitive closure



Definition

Let $R \subseteq X \times X$ be a homogeneous relation on X . A **transitive closure** of the relation R is the minimum (in sense of inclusion) transitive relation $\overline{R}$, which extends R , i.e. $R \subseteq \overline{R}$.

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It is enough to take $\overline{R} = \bigcap_Q Q$, an intersection of all transitive extensions $Q \supseteq R$. The total relation $X \times X$ is transitive.



Transitive closure

Definition (Reminder)

A digraph without (directed) cycles is called a **dag** (Directed Acyclic Graph).

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Given dag $G = (V, A)$ it is easy to create a poset, because $\bar{A}$ is a strict order.



Cover graph

Definition

Let $(X, \preceq)$ be a poset. We say that $x \in X$ **covers** $y \in X$, and denote $x \triangleleft y$, if $x \prec y$ and there is no $z \in X$ such that $x \prec z \prec y$.

Graph $(X, \triangleleft)$ is called a **cover graph** of poset $(X, \preceq)$.

Proposition

Let $(X, \preceq)$ be a finite poset and $(X, \triangleleft)$ — its cover graph. Then there is $\overline{\triangleleft} = \prec$.

Finiteness is important because e.g. $(\mathbb{Q}, \leq)$ has an empty cover relation, although for $(\mathbb{N}, \leq)$ it is OK (Peano axioms).



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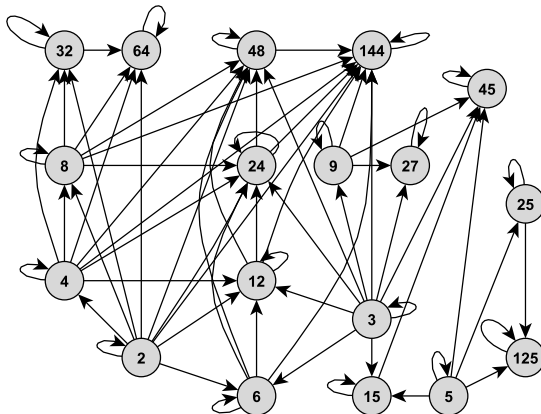
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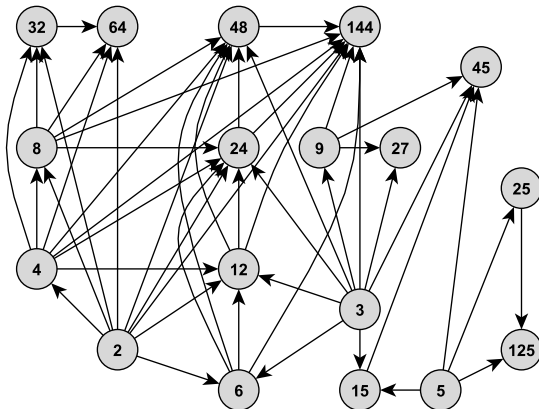
A cover graph is triangle free.

Let's draw the poset



1. The original poset

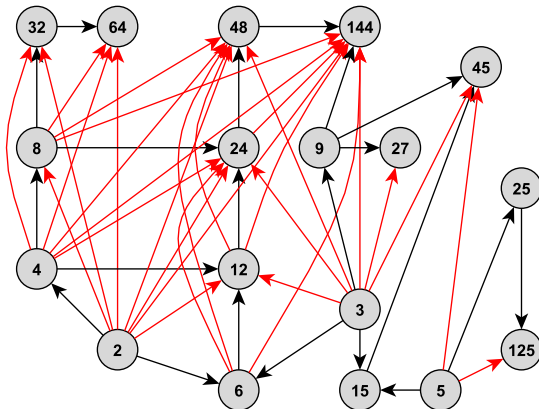
Let's draw the poset



2. The strict order (loopless)



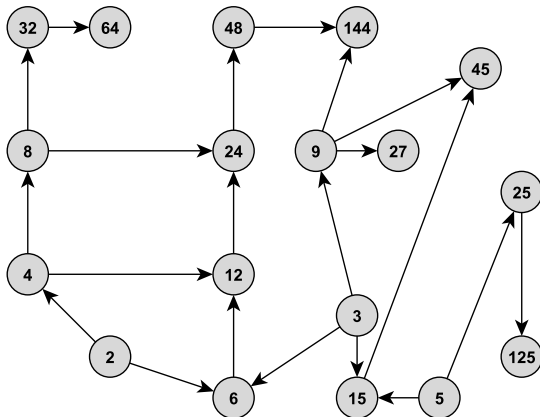
Let's draw the poset



3. The strict order (loopless), red arcs forced by transitivity



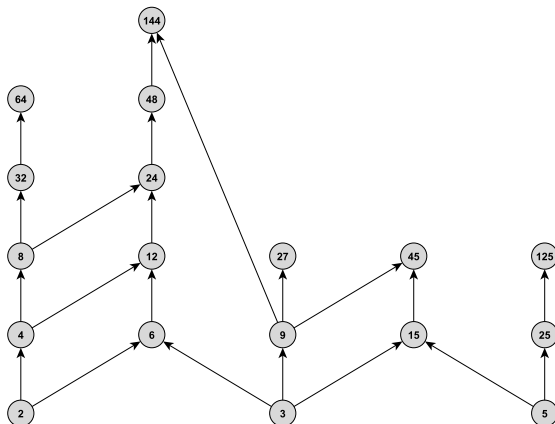
Let's draw the poset



4. The cover graph

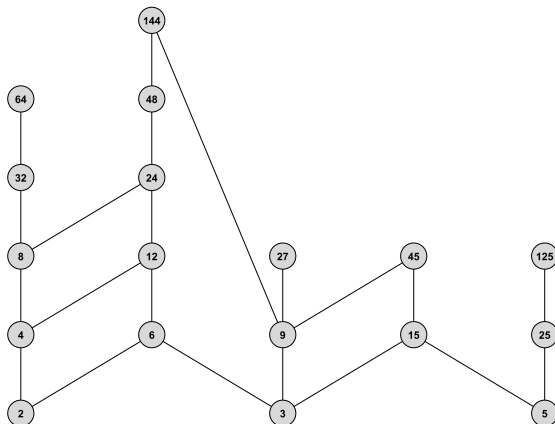


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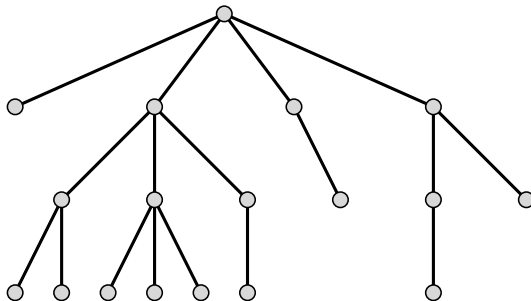
5. The cover graph, arrows up

Hasse diagram



6. The **Hasse diagram** — a drawing of cover graph without arrows — their direction is determined by the levels of the vertices.

Hasse diagram

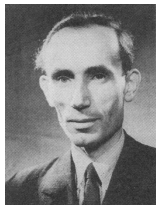


Remark: A rooted tree.

Comparability graphs

Definition

Let $(X, \preceq)$ be a poset. We say that elements $x, y \in X$ are **comparable** when $x \preceq y$ or $y \preceq x$ holds. Let us denote them $x \sim y$ and, when they are **incomparable** $x \perp y$.



One can characterise (Gallai [1967]) which graphs (simple, undirected) can be **comparability graphs**, i.e. graph of the form (X, E) , where

$$E = \{\{x, y\} : x, y \in X \wedge x \neq y \wedge x \sim y\}.$$

Question

Can graph C_{2n} be a comparability graph?
For which $n \in \mathbb{N}$, graph C_{2n+1} can be that?



Chains, antichains, width and height

Definition

Let $(X, \preceq)$ be a poset. Set $L \subseteq X$ is called a **chain**, when elements of L are pairwise comparable. Set $A \subseteq X$ is called **antichain**, when elements of A are pairwise incomparable.

Of course, the order $\preceq$ truncated to a chain is a total order.

Definition

Let $(X, \preceq)$ be a poset.

The **length** of $(X, \preceq)$ is the number $\ell(X, \preceq) = \max_{L - \text{a chain}} |L|$.

The **width** of $(X, \preceq)$ is the number

$\mathcal{W}(X, \preceq) = \max_{A - \text{an antichain}} |A|$.



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$\mathcal{W}(X, \preceq) = \max_{A - \text{an antichain}} |A|$.



Dilworth's theorem

Theorem (Dilworth [1950])

The width of a finite poset $(X, \preceq)$ is equal to the minimum number of chains that cover all elements of X .



Dilworth's theorem

Theorem (Dilworth [1950], different formulation)

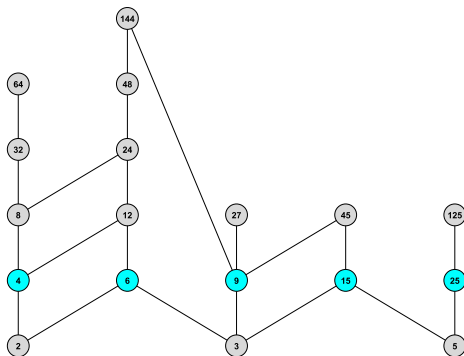
For finite poset $(X, \preceq)$ there is $\mathcal{W}(X, \preceq) = \mathcal{V}(X, \preceq)$, where $\mathcal{V}(X, \preceq) = \min_{\mathcal{L}} |\mathcal{L}|$, $\mathcal{L} = \{L_i: L_i \text{ is a chain}\}$, $\cup_{L \in \mathcal{L}} L = X$.

Proof sketch.

The inequality

$$\mathcal{W}(X, \preceq) \leq \mathcal{V}(X, \preceq)$$

is obvious, since no chain can contain 2 elements of antichain.





Dilworth's theorem

Theorem (Dilworth [1950], different formulation)

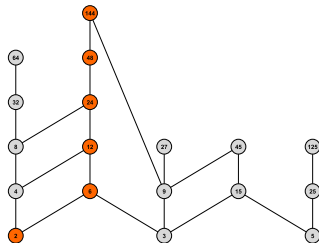
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Proof sketch.

We show $\mathcal{W}(X, \preceq) \geq \mathcal{V}(X, \preceq)$ by induction on $n = |X|$. For $n \leq 1$ — it is obvious. Suppose now that the inequality holds for every n -element poset, where $n < n_0$.

Let $(X, \preceq)$ be a poset such that $|X| = n_0$, $w = \mathcal{W}(X, \preceq)$. Let L be the longest chain in $(X, \preceq)$.

If $\mathcal{W}(X \setminus L, \preceq_{|X \setminus L}) < w$ holds, then OK, the inductive assumption implies the thesis.





Dilworth's theorem

Theorem (Dilworth [1950], different formulation)

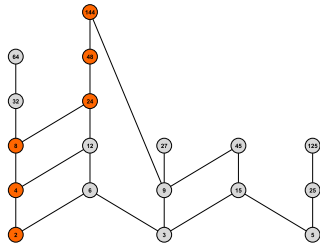
For finite poset $(X, \preceq)$ there is $\mathcal{W}(X, \preceq) = \mathcal{V}(X, \preceq)$, where $\mathcal{V}(X, \preceq) = \min_{\mathcal{L}} |\mathcal{L}|$, $\mathcal{L} = \{L_i: L_i \text{ is a chain}\}$, $\cup_{L \in \mathcal{L}} L = X$.

Proof sketch.

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If $\mathcal{W}(X \setminus L, \preceq_{|X \setminus L}) < w$ holds, then OK, the inductive assumption implies the thesis. Suppose then otherwise, that $\mathcal{W}(X \setminus L, \preceq_{|X \setminus L}) = w$.





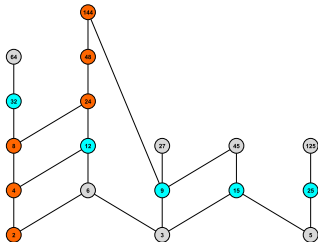
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Proof sketch.

Let $A = \{a_1, a_2, \dots, a_w\}$ be an antichain in $(X \setminus L, \preceq|_{X \setminus L})$. Let $B = \{x \in X: \exists_{a \in A} x \preceq a\}$, $T = \{x \in X: \exists_{a \in A} a \preceq x\}$. Then $B \cup T = X$, $B \cap T = A$ and $|B|, |T| < n_0$, since $\max L \notin B$, $\min L \notin T$. From the inductive assumption both B and T are covered by w chains: $B = B_1 \cup B_2 \cup \dots \cup B_w$, $T = T_1 \cup \dots \cup T_w$. Let us number them so that $a_i \in B_i \cap T_i$ (it is possible because each element of A belongs to exactly one chain B_i and one T_i). Then $L_i = B_i \cup T_i$ is covering of X by w chains. Thus the theorem holds by induction.





Dilworth's theorem

Theorem (Dilworth [1950])

The width of a finite poset $(X, \preceq)$ is equal to the minimum number of chains that cover all elements of X .

Corollary (The second Dilworth's theorem)

The width of a finite poset $(X, \preceq)$ is equal to the minimum number of disjoint chains that cover all elements of X .

It is enough to remove from the chains the repeating elements.



Dilworth's theorem

Theorem (Dilworth [1950])

The width of a finite poset $(X, \preceq)$ is equal to the minimum number of chains that cover all elements of X .

Theorem (Mirsky [1971], dual theorem)

The length of a finite poset $(X, \preceq)$ is equal to the minimum number of antichains that cover all elements of X .

The proof is (relatively) easy — extra exercise.



Dilworth's theorem

Theorem (Dilworth [1950])

The width of a finite poset $(X, \preceq)$ is equal to the minimum number of chains that cover all elements of X .



Corollary (Theorem of Erdős, Szekeres [1935])

Given $r, s \in \mathbb{N}$ every finite sequence of different $rs + 1$ real numbers contains either an increasing subsequence of length $r + 1$ or a decreasing one of length $s + 1$.



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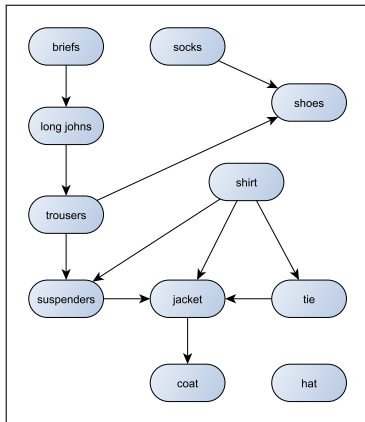


Corollary (Theorem of Erdős, Szekeres [1935])

Given $r, s \in \mathbb{N}$ every finite sequence of different $rs + 1$ real numbers contains either an increasing subsequence of length $r + 1$ or a decreasing one of length $s + 1$.

Proof. Let $Z = (z_1, \dots, z_{rs+1})$. Let us define the relation $\prec$ on index set $X = \{1, 2, \dots, rs+1\}$ by $i \prec j \Leftrightarrow i < j \wedge z_i < z_j$. This is strict order, hence $(X, \preceq)$ is a poset. Chains define increasing subsequences of Z , while antichains — decreasing ones. Either there is an antichain of length $s + 1$, or X is covered by s chains. Then, by the pigeonhole principle, one of them must have at least $r + 1$ elements.

How to dress an (indecisive) mathematician?



cf. Cormen et al., 2009

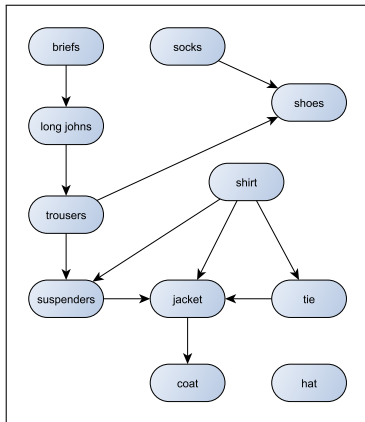
Problem

Given the order of tasks, schedule them.

This is possible provided that the directed graph indicating the order is acyclic.

How to determine the schedule?

How to dress an (indecisive) mathematician?



cf. Cormen et al., 2009

Problem

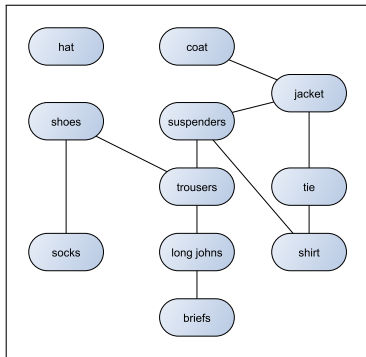
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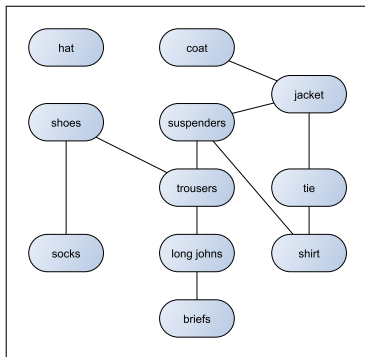
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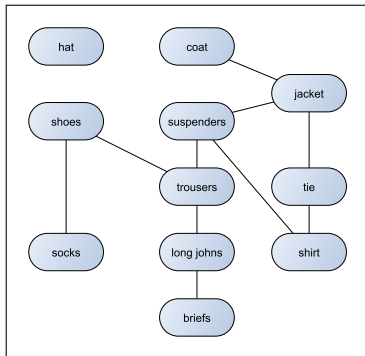
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In this example, eg.: briefs, socks, shirt, tie, long johns, trousers, suspenders, jacket, coat, shoes, hat; or hat, shirt, tie, socks, briefs, long johns, trousers, shoes, suspenders, jacket, coat.



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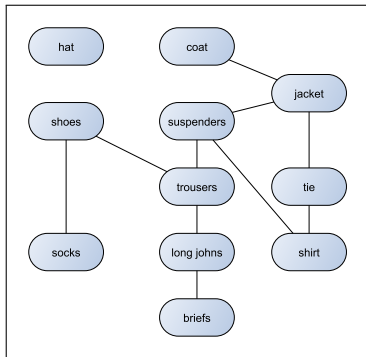
How to determine the schedule?

Definition

Let $\preceq$ be a partial order of set X . We call $\leq$ an **linear extension** of $\preceq$, when $\leq$ is a total order of X and $\preceq \subseteq \leq$.



How to dress an (indecisive) mathematician?



Problem

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Proposition

Every partial order has a linear extension.

For finite sets — a simple induction. For infinite sets the proof requires the axiom of choice (Marczewski/Szpilrajn [1930], earlier Banach, Kuratowski, Tarski).